

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

In this digital era, many people have come to rely on the Internet for everything, making it a necessity for them. People often share sensitive data, such as passwords, bank details, personal information and other private information that only the individual knows (Ahmad et al., 2024). Attackers have developed new methods to target individuals, largely because most people rely heavily on digital tools. Phishing is the most common and harmful type of cyberattack.

Phishing attacks have changed from standard email fraud into a major cybersecurity risk that includes SMS-based fraud (smishing) and malicious websites (Saranya et al., 2025). Attackers take advantage of user weaknesses, tricking them into revealing sensitive information via fake URLs, fraudulent messages and malicious websites (Popescul et al., 2025). One of the methods that attackers usually use is pretending to be someone the user knows and trusts, and tricking the user into giving them their sensitive information.

Previously, phishing attacks, such as emails, were easier to detect because the errors and design were too obvious. Then, using traditional methods, a blacklist is used to stop the attack. However, because phishing methods are more sophisticated, they have surpassed conventional, rule-based security measures, requiring more advanced solutions. Thus, the traditional cybersecurity methods cannot adapt to new phishing techniques due to their inherent limitations. Therefore, it is clear that the industry demands advanced and adaptive detection systems.

The majority of cybersecurity experts have shifted to Artificial Intelligence (AI) and Machine Learning. For traditional Machine Learning, a lot of manual work is required to identify key data features (Alshingiti et al., 2023). Meanwhile, handling high-dimensional data with many irrelevant features can slow down the system and reduce classifier accuracy.

The limitations of current technology reveal a significant gap, which is fast systems, power systems that can manage different kinds of data at the same time and provide accurate results. To address this, the study introduces a back-end engine that applies Binary Harris Hawks Optimization (BHHO) to automate feature selection. The engine focuses on the most relevant features, reduces manual workload and eliminates redundant data. This allows Machine Learning to operate more efficiently and accurately across different communication channels.

1.2 Problem Statement

Current Machine Learning-based phishing detection models frequently process a large number of features from datasets without proper optimization (Alshingiti et al., 2023). As a result, the problem can cause irrelevant and redundant features, thus introducing noise into the data, which leads to increased computational costs, potential overfitting and influences the model accuracy (Danasingh et al., 2020).

Feature selection is a fundamental step in machine learning preprocessing, as it improves the model's performance, reduces computational cost, and makes the model easier to understand by focusing on the most useful input features (Mohammed & El-Feky, 2025). Furthermore, feature selection techniques are commonly categorized into three main types, which are filter-based methods, wrapper-based methods and embedded methods. Firstly, the filter-based method is an independent technique that selects the important features using statistical ranking such as information gain, chi-

square test, Fisher's score and correlation coefficient without relying on any learning algorithms (Kaur et al., 2021). Next, according to Kaur et al. (2021), the wrapper-based method selects the best set of features by testing directly with learning algorithms using search strategies such as sequential selection and heuristic search. Lastly, the embedded method performs feature selection during the training, combining selection and learning in one process to improve efficiency and accuracy.

Although the filter-based methods are effective, these methods usually keep a set of features that may not be the best for classification, while wrapper-based methods are computationally demanding because they depend on the repeated training of learning models (Mohammed & El-Feky, 2025). For embedded methods, the selected features are biased towards one specific model, even though reduced risk of overfitting (Syeda & Asghar, 2023). These limitations show that the existing feature selection methods may not sufficiently handle high-dimensional data and complex datasets. Therefore, it is essential to develop a feature selection model that enhances model accuracy and reduces computational costs.

1.3 Research Objectives

The main objective of this study is to investigate the ability of the proposed feature selection for phishing detection. To achieve this goal, the following objectives are outlined:

1. To identify the important features for phishing detection using Binary Harris Hawks
2. To compare the performance of Binary Harris Hawks with the traditional feature selections.
3. To evaluate the effectiveness of the selected features in improving phishing detection accuracy using machine learning classifiers.

1.4 Research Scope

The scope of this research focuses on developing phishing detection system that is capable on analyze three communication channels, which are emails, URL and SMS. The study focuses on feature selection process using Binary Harris Hawks Optimization (BHHO) to identify the most relevant features for phishing detection. Data preprocessing and classification are performed to evaluate the effectiveness of the selected features. In addition, publicly available datasets that contain labeled phishing and legitimate samples are being used without monitoring live network traffic or developing a user interface.

1.5 Research Significance

This study contributes to academic knowledge by analyzing how optimized feature selection can enhance phishing detection. According to Savyanavar et al. (2024), the study highlights several weaknesses of traditional feature selection methods, which include irrelevant or redundant data, thus offering valuable guidance for future research in phishing detection and data preprocessing. The findings can be a reference to other researchers who seek more efficient feature selection methods in cybersecurity systems.

Additionally, this study improves the efficiency and accuracy of phishing detection systems, which focus on the most important features while reducing unnecessary data and computational costs (Danasingh et al., 2020). The proposed approach can be applied in real-world detection systems through emails, URL and SMS that can help organizations to develop more robust and effective cybersecurity solutions.

CHAPTER 2

2.0 LITERATURE REVIEW

2.1 Overview of Phishing

Phishing is a cyberattack where attacker pretend to be reliable organizations to obtain sensitive details like usernames, passwords or financial information. It can take place across different channels such as email, URL or text messages (Ahmad et al.). Attackers often create phishing materials and websites that mimic like a legitimate one to build user trust and trick individuals into sharing their information (Joseph & Srinivasan, 2025). Therefore, the complicated design of fake links and interfaces make it hard for users to detect threats unless they have enough awareness or defense mechanism (Ahmad et al., 2024).

The main goal of phishing attacks is to take advantage of human weaknesses and psychological patterns to trick users into revealing private information (Korkmaz, 2026). Furthermore, phishing methods have become more advanced and complex over time, making them difficult to detect with conventional security techniques (Ahmad et al., 2024). Conventional detection methods such as blacklists and rule-based systems depend on static repositories of attack signatures and predefined rules (Doshi et al., 2025). As a result, these methods find it challenging to adjust to modern approaches and zero-day threats which highlights the urgent need for detection systems that are adaptive, intelligent and automated (Saranya et al., 2025).

2.2 Types of Phishing Attacks

2.2.1 Email-Based Phishing Attacks

Email phishing refers to the act of sending fraudulent messages via email that come from reliable sources such as banks, service providers and coworkers, in which to trick the users into clicking a malicious link or giving their sensitive information like credit card numbers or passwords (Ramprasath et al., 2023). Emails are commonly used with persuasive language, urgent messages and inserted links or files to manipulate users (Chanis & Arampatzis, 2024). Phishing emails can be detected through features such as suspicious sender details, mismatched domains, abnormal subject headings and language patterns in the message (Zhang et al., 2025). Choosing relevant email features is a must to enhance the accuracy of the phishing detection model.

2.2.2 SMS-Based Phishing Attacks

SMS based phishing, also known as smishing, is an attack that target mobile phone of the user through text messages in which the text message can pretend to be a legit message. The text messages commonly contain shortened links or urgent messages to triggers user in act quickly (Mishra & Soni, 2020). The previous research shows that short text messages in short forms or symbol forms and text messages consists of idioms and abbreviations are characteristics of the smishing attacks (Mishra and Soni, 2021). Smishing attacks are difficult to detect because of the feature limitations and the lack of sufficient real datasets.

2.2.3 URL-Based Phishing Attacks

Attackers use malicious links to redirect users to fraudulent sites that are designed exactly like legitimate platforms (Ewasakul & Oransirikul, 2024). The links can be spread through emails, SMS or online advertisements. Moreover, detecting URL phishing involves analyzing features such as the length of the URL, special characters, suspicious term and abnormal domain patterns (Kustiawan & Ghauth, 2025). As phishing URLs change rapidly, choosing the most important features is vital for enhancing detection accuracy and reducing complexity.

2.3 Types of Feature Selection

Feature selection techniques are commonly categorized into three types, which are filter-based methods, wrapper-based methods and embedded methods. Each type of feature selection methods has different characteristics, advantages and limitations that will be discussed below.

2.3.1 Filter-based methods

Based on Mandal et al. (2021), filter-based methods are a technique that selects the most important features from the datasets without relying on other classification models. The key features are chosen based on the statistical significance to support the classification or prediction of the data. One of the advantages of filter-based methods is in text classification and sentiment analysis, where dimensionality is reduced by ranking and choosing the most significant terms to enhance both accuracy and clarity (Mohammed & El-Feky, 2025). The limitation of this model is assuming that the important features do not change between datasets, which reduces adaptability in real-world applications (Mohammed & El-Feky, 2025).

2.3.2 Wrapper-based methods

According to Yin et al. (2023), wrapper-based methods are a technique that evaluates feature subsets based on actual performance in a given machine learning model. Wrapper methods are not like filter methods that are independent, it can rely on different classifications or prediction models. Each of the selected feature subsets requires the model to train and test, which results in higher computational costs and longer processing time. The advantage of using a wrapper-based method is that the method considers how features depend on each other while interacting with the classifier and it is not highly demanding in terms of computation (Kaur et al., 2021). As for limitation is, methods carry the risk of overfitting but deliver good results and require more time to complete. (Kaur et al., 2021).

2.3.3 Embedded methods

Embedded methods select useful features during training, which combines selection and learning in a single process to enhance efficiency and accuracy (Kaur et al., 2021). The advantages of embedded methods are that the methods can perform variable selection, feature selection and classification together during the training process (Kaur et al., 2021). This method depends on a classifier, which can result in different feature selection based on the classifier's assumptions (Syeda & Asghar, 2023).

2.4 Machine Learning Approaches

Random Forest (RF) is commonly applied in phishing detection because its ensemble of decision trees enhances accuracy and minimizes overfitting by combining multiple predictions. It can efficiently process URL-based features without requiring web content access and when paired with deep learning techniques such as CNN for

feature extraction, it achieves accuracy levels exceeding 99% (Yang et al., 2021). According to Kavya & Sumathi (2024), RF is also applied to structured feature data such as URL metadata and domain details, enabling a comprehensive analysis of phishing indicators throughout the process. Random Forest is considered a suitable option for intelligent phishing detection systems, as it offers stronger generalization and faster execution compared to classifiers like ANN, SVM, C4.5, and Rotation Forest (Keerthana et al., 2025).

Furthermore, XGBoost or Extreme Gradient Boosting is an ensemble technique that combines multiple weak learners to form a stronger model, thus enhancing predictive performance (Gu & Xu, 2022). Based on Saraswathi et al. (2024), XGBoost can manage large datasets and complex patterns effectively making it a reliable option for phishing detection. It also often producing excellent results in classification tasks.

In addition, K-Nearest Neighbors (KNN) is a simple algorithm that uses training data to classify unseen data based on a chosen distance measure (Gu & Xu, 2022). According to Kalikiri et al. (2025) KNN is a non-parametric method that categorizes data points based on how they are to one another, using majority vote among the nearest neighbors. It also often used to fill in missing values in datasets. The main idea is that points located near each other in the feature space are likely to be similar. Therefore, the algorithm predicts the class or value of a new point by looking at the attributes of its closest neighbors.

Moreover, Support Vector Machines (SVM) creates boundaries in high-dimensional space to separate different classes while keeping the widest possible margin between them. It works effectively when the classes are clearly different in their features making it useful for separating phishing URL from legitimate ones (Saraswathi et al., 2024). According to Ajibade et al. (2025), SVM are efficient in memory use because only the support vectors affect the decisions. They work well in high-dimensional spaces, even when there are more features than samples. However, they face difficulty with noisy or overlapping data and can become slow when

handling large datasets. The accuracy also depends on choosing the right kernels and parameters so poor choices can reduce performance.

2.5 Related Works

2.5.1 Existing Methods of Feature Selections

Previous research has employed different feature selection methods to enhance phishing detection across emails, SMS, URLs, websites and online services. Mat Rani et al. (2023) applied Information Gain and TreeSHAP, highlighting that XGBoost achieved the highest accuracy of 98.59%, though its effectiveness diverse depending on the dataset. Also, Bahaghighat et al. (2023) used MOTEENN and feature reduction methods where XGBoost achieved 99.2% accuracy. Their work also highlighted the importance of large phishing datasets and more advanced deep learning approaches.

Dahr and Gaafar (2025) stated that using Chi-Square feature selection with XGBoost gave 75.76% accuracy for smishing and intrusion detection although some main features might be lost during process. Connolly and Atlam (2025) presented a hybrid feature selection combined with stacking ensembles, where XGBoost reached 99.53% accuracy and 99.85% recall in detecting email phishing. However, they pointed out that the variety of datasets remains a limitation.

Other researchers have discovered optimization and ensemble-driven feature selection methods. Siva et al. (2024) obtained strong accuracy results through canopy-based feature selection combined with hybrid classifiers. In another research, Savyanavar et al. (2024) reported 96.57% accuracy using ensemble feature reduction, but emphasized that technical detection alone may not be enough to counter social engineering tactics. Similarly, Kocyigit et al.

(2024) used a Genetic Algorithm to simplify features for URL-based phishing detection, which improved efficiency but exclude the HTML-related features.

In general, current approaches enhance detection accuracy but face challenges including reliance on specific datasets, potential loss of important features, and restricted adaptability. These issues point to the requirement for stronger and more advanced feature selection strategies.

Table 2.1 Comparisons between feature selection methods

Author(s), Year	Types of Phishing	Feature Selection Method	Classifier used	Main Findings	Limitations
Mat Rani et al. (2023)	Email, spear phishing, whaling, voice, and website phishing	Information Gain (IG) and TreeSHAP	Naïve Bayes (NB), Random Forest (RF), and XGBoost	XGBoost achieved the highest accuracy of 98.59%	Naïve Bayes underperformed on Dataset 2; performance is highly dataset-dependent
Bahaghighat et al. (2023)	E-banking, E-commerce, social media	SMOTEENN & dropping constant features	LR, KNN, NB, RF, SVM, XGBoost	XGBoost reached 99.2% accuracy	Needs more phishing samples & Deep Learning
Dahr & Gaafar (2025)	SMS Spam (Smishing) and Network Intrusion (DDoS)	Filter (Chi-Square), Wrapper (Lasso, RFE),	XGBoost	Filter (Chi-Square) was the most accurate (75.76%)	Feature selection can lead to a loss of features; models may be weak for datasets containing causal features
Connolly & Atlam (2025)	Email Phishing	Hybrid feature selection	Stacking Ensemble: RF and	Achieved 99.53% accuracy	Standard public datasets lack attachment

			XGBoost (base learners) with Logistic Regression (meta learner)	and 99.85% recall	samples, limiting real-world diversity
Siva et al. (2024)	Fake emails and fraudulent websites	Canopy feature selection with grid search and cross-fold validation	DT, LR, RF, NB, GB, KNN, SVM, and a Hybrid LSD model (LR+SVM+DT)	The Hybrid LSD and Stacking Classifiers achieved 100% accuracy	Previous methods relied on single anti-spam services and struggled with zero-hour attacks
Savyanavar et al., 2024	Email, Smishing, Vishing, Whaling	Ensemble reduction (IG, Chi-Square, Gain Ratio, Correlation)	Random Forest	Achieved 96.57% accuracy	Psychological manipulation is noted as a factor that technical detection alone may not fully stop
Kocyigit et al., 2024	URL-based phishing	Genetic Algorithm (GA) with Local Optimization	Random Forest, XGBoost, and ANN	GA reduced features to fewer than 45	Timing of feature calculation was not factored into the current work; future needs to add HTML content

2.6 Summary

On this chapter review the main concepts and existing research related to phishing detection systems. It starts with an overview of phishing, explain the purpose and how phishing become more sophisticated over time. Different phishing channels such as email, SMS and URLs were described to show the variety of techniques used by

attackers and the challenges in detecting them. Then, discussed about feature selection methods, covering filter-based, wrapper-based and embedded approaches. Each of the method was outlines with its advantages and weaknesses especially in handling large and high dimensional datasets. The section highlighted that selecting appropriate features is crucial for improving detection accuracy and minimizing redundant data. The chapter also looked at several machine learning algorithms that are widely applied in phishing detection, including Random Forest, XGBoost, K-Nearest Neighbors, and Support Vector Machines. The discussion focused on how these algorithms function and how effective they are in distinguishing phishing and legitimate data. In conclusion, previous research on feature selection for phishing detection were examined and compared. While many of the existing methods delivered strong results, they continue to encounter challenges such as depend on particular datasets, the loss of key features, and restricted flexibility. These shortcomings emphasize the importance of developing a more effective feature selection strategy, which is examined in the proposed study.

CHAPTER 3

3.0 RESEARCH METHODOLOGY

3.1 Introduction

This study applies a quantitative research approach because it focuses on the data that can be measured and expressed through numbers. It evaluates performance using metrics such as accuracy, precision, recall and F1-score. The study can provide a transparent evaluation of the hybrid model and traditional methods. Additionally, the experimental research is also applied where the models are trained, tested and evaluated systematically in a monitored environment using datasets.

3.2 The Methodology Framework

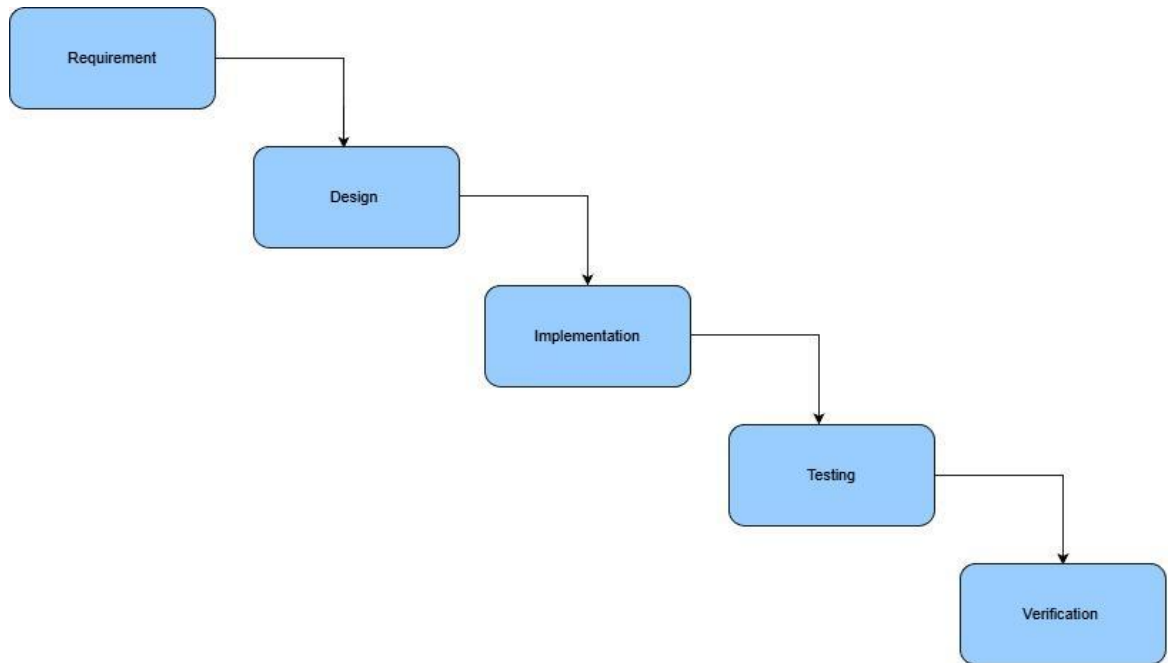


Figure 3.1 The Waterfall Model

In the requirement phase, objectives, scope and limitations of the research is identify and defined. Many researchers have set a clear objective for detecting phishing in emails, URL and SMS and emphasize non-functional requirements such as improved accuracy and efficiency (Siva et al., 2024). Based on Mughaid et al. (2022), URL-based systems evaluate large datasets using performance metrics accuracy, precision, recall and F1-score. Meanwhile, email-based system needs to classify whether the email is phishing or legitimate. Email-based system uses the same metrics but include the false-positive rate.

The overall system structure is planned in the design phase. This involves designing a data preprocessing workflow. The choice of feature selection methods and classification techniques is specified to ensure the system fulfills the requirements. This phase defines the workflow for cleaning and transforming raw data, identifying

relevant features, enhancing model performance and selecting algorithms (Kadulkar et al., 2022).

This implementation phase focuses on implementing the system based on design requirements. The selected software tools are used to implement preprocessing data, feature selection attributes and development of models such as Random Forest, Support Vector Machine, K-Nearest Neighbors, Artificial Neural Network, Deep Neural Network, Convolutional Neural Network, XGBoost and hybrid ensembles on phishing datasets (Mughaid et al., 2022).

Performance evaluation of the implemented system is conducted through testing using metrics measures such as accuracy, precision, recall and F1-score through different algorithms and feature sets (Gualberto et al., 2020). Multiple configurations are tested to verify systems reliability and to achieve the research objectives.

In this verifications phase is ensure that the development system meets the predefined requirements and research objectives. The outcomes are analyzed and validated by comparing with existing methods. This phase ensures that the proposed solution is effective, consistent, and relevant to phishing detection (Gualberto et al., 2020).

3.3 System Diagram

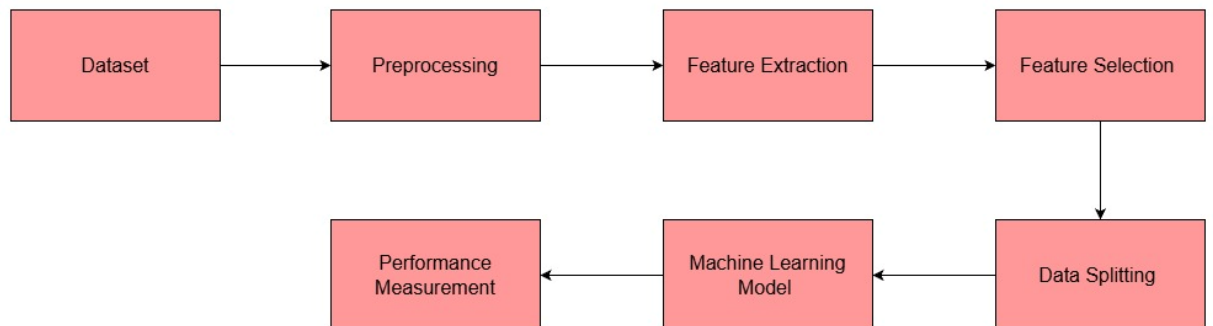


Figure 3.2 The System Workflow

The workflow of this phishing detection system starts with the collection of datasets from three different sources, which are email, SMS and URL datasets. These datasets contain phishing data and legitimate samples to be an input to the system. Phishing and legitimate samples are collected from UCI Machine Learning Repository, URL-Phish and publicly available text message. The PhiUSIIL phishing URL dataset from the UCI Machine Learning Repository comprises 235,795 data points, consisting of 134,850 legitimate URLs and 100,945 phishing URLs. Also, 54 extracted features represent lexical and structural properties (UCI Machine Learning Repository). The URL-Phish dataset contains 111,660 unique URLs with 22 extracted lexical and structural features and was collected from PhishTank and various trusted sources (Mendeley Data). The structural features of this dataset are URL length, domain length, number of subdomains, digit ratio, entropy, and HTTPS usage. The SMS phishing dataset is compiled from publicly available text messages. This dataset has 5971 text messages labeled as Legitimate or Smishing. It includes 489 spam messages, 638 smishing messages, and 4844 ham messages.

Data preprocessing is the phase where raw data need to clean and normalize before continue with further analysis. The first step is converted to lowercase to remove case sensitivity issues. Punctuations marks, numbers, special characters and unnecessary symbols which are irrelevant data are then removed to reduce noise. Next, tokenization divides the URL or email text into meaningful components such as domain names, path segments or words allowing feature extraction methods like TF-IDF to identify important textual patterns. Additionally, stemming or lemmatization convert word to their root forms and stop-word removal to eliminate common but uninformative words especially when analyze email content. These preprocessing operations improve data consistency, reduce dimensionality and enhance the quality of features used for the following stages.

Feature extraction is performed using Term Frequency-Inverse Document Frequency (TF-IDF) technique once the data has been preprocessed. TF-IDF convert the cleaned textual data into numerical feature vectors by assigning weights to words based on their importance within a document and across the whole dataset. The TF-IDF representation highlights characteristic phishing keywords like 'verify,' 'urgent,' and

'account suspended,' which are common in phishing emails but appear less frequently in legitimate messages. For phishing website, character-based TF-IDF features extracted from noisy HTML code and plain text have proven effective in detecting subtle patterns without relying on external services, thereby improving the identification of zero-hour phishing attacks with similarly high accuracy. Similarly, applying TF-IDF to login URLs captures lexical differences that separate phishing sites from genuine login pages, addressing real-world scenarios where URLs may appear visually similar but carry malicious intent. Thus, each document is encoded as a high-dimensional vector that indicates the relative significance of its textual elements.

Binary Harris Hawks Optimization (BHHO) is applied as the feature selection method. BHHO takes inspiration from the cooperative hunting strategy of Harris hawks and is designed to identify an optimal set of features within a binary feature space. In this process, each hawk corresponds to a possible feature subset obtained from the TF-IDF features, where a value of one represent that a feature is chosen and a value of zero show that is not included. During optimization, an initial group of hawks is generated at random, and each one is assessed using a fitness function that accounts for both classification accuracy and the number of chosen features. The algorithm changes between exploration and exploitation phases to effectively search the feature space. In the exploration stage, BHHO tests various feature combinations to prevent early convergence, while in the exploitation stage, it concentrates on refining promising subsets to enhance solution quality. Through repeated updates, BHHO slowly moves toward an optimal subset of features that boosts detection performance while reducing dimensionality. After the optimal feature subset is obtained, the reduced set of features is passed to the next stages of the workflow. The optimized features are then split into training and testing datasets through the data-splitting process. These datasets are used to train and evaluate the machine learning classifier. Integrating BHHO into the workflow ensures that only the most relevant and informative features are applied for classification, which improves accuracy, reduces overfitting, and enhances computational efficiency.

After the process of feature selection using BHHO, the dataset is divided into training and testing sets. According to Mohamad Asraf Daniel et al. (2025), the standard ratio is 80:20 and 60:40, which are used to balance training and evaluation phases. The training datasets contains varied set of labelled examples comprising both legitimate and phishing. The purpose of the datasets is to record the various characteristics and patterns that are linked to phishing attacks. Then, the testing datasets are structured to measure the ability of the model in handling unfamiliar data. This process ensures that the model performs consistently, reduces overfitting and offers a better measure of phishing detection.

After data splitting, the machine learning model process is introduced. Random Forest algorithm functions as a classification model rather than a feature extraction method. Before reaching the classifier, all textual data are already preprocessed and converted into numerical form using the TF-IDF method. The Random Forest then uses these feature vectors along with their class labels to learn how to separate phishing from legitimate cases during training. It builds multiple decision trees from random subsets of the data, with each tree developing its own rules based on different combinations of TF-IDF features to distinguish phishing data from legitimate samples. Each decision tree produces a separate classification when performing a prediction. The final outcome is decided through majority voting across the trees, and this combine strategy strengthens reliability while improving the accuracy of phishing detection.

Lastly, the performance of the phishing detection system is assess using standard classification metrics. A confusion matrix is generated to outline prediction results in terms of true positives, true negatives, false positives, and false negatives. Based on these values, evaluation metrics including accuracy, precision, recall, and F1-score are calculated. Accuracy reflects the overall correctness of the predictions, precision represented the proportion of phishing instances correctly identified, recall indicates the system's ability to detect actual phishing cases, and the F1-score provides a balanced measure of precision and recall. Employing multiple evaluation metrics ensures a comprehensive assessment of the system's performance, particularly when dealing with imbalanced datasets.

3.4 Performance Evaluation Metrics

To evaluate the effectiveness of the proposed Binary Harris Hawks Optimization (BHHO) and performance outcomes, four standard evaluation metrics will be applied. The metrics serve as quantitative measures of how effectively the model can differentiate phishing attacks and limit the false alarms. According to Alshingiti et al. (2023), the prediction results were assessed using accuracy, precision, recall and confusion matrix to evaluate overall performance.

- Accuracy: The accuracy is the percentage of correct predictions, showing how well the system identifies both phishing and legitimate cases (Alshingiti et al., 2023).

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP}$$

- Precision: The precision is the number of phishing webpages that are correctly identified as phishing (Alshingiti et al., 2023).

$$Precision = \frac{TP}{TP + FP}$$

- Recall: Recall is evaluated as how effectively the model detects and identifies all genuine phishing threats present in the dataset (Alshingiti et al., 2023).

$$Recall = \frac{TP}{TP + FN}$$

- F1-Score: Process of calculating the harmonic mean of precision and recall to form one unified measure (Alshingiti et al., 2023).

$$F1 - Score = \frac{2 \times (Precision \times Recall)}{(Precision + Recall)}$$

3.5 Experimental Setup

3.5.1 Hardware Requirements

The computer system used to run the experiments is powered by an AMD Ryzen 7 5825U processor. This CPU features an 8-core, 16-thread architecture with a maximum boost clock of 4.5 GHz. The system is equipped with 12GB of DDR4 memory and utilizes AMD Radeon™ Graphics for integrated visual and parallel processing tasks.

Table 3.3 Hardware Requirements

No.	Hardware Components	Specification
1	System Model	HP Laptop 15-fc0xxx
2	Processor (CPU)	AMD Ryzen 7 5825U
3	Cores / Threads	8 Cores / 16 Logical Processors
4	Base Clock Speed	2.0 GHz (Max Boost up to 4.5 GHz)
5	Memory (RAM)	12 GB DDR4 3200MHz
6	Storage	512 GB NVMe SSD
7	Graphics (GPU)	AMD Radeon™ Graphics (Integrated)

3.5.2 Software Requirements

This section is about software requirements used in developing a phishing detection system. Below are the listed tools and platforms that are widely used in this project and are suitable for the back-end system.

Table 3.4 Software Requirements

Category	Software / Tool	Purpose
Operating System	Windows 10 / Windows 11	A platform used to develop and run the phishing detection system
Programming Language	Python 3.	Used for data preprocessing, feature selection, and classification
Development Environment	Anaconda Distribution	Manages Python environments and required libraries
Interactive Environment	JupyterLab	Used for data analysis, experimentation, and result visualization
Data Processing Libraries	NumPy, Pandas	Used for numerical operations and dataset handling
Machine Learning Library	Scikit-learn	Used for model training, testing, and evaluation
Visualization Libraries	Matplotlib, Seaborn	Used to visualize results and performance metrics
Dataset Format	CSV	Used for storing and processing phishing datasets

3.6 Gantt Chart

A Gantt chart was created to show the project progression across 28-week which is around one year in an academic semester. This is to ensure that every phase, starting from the introduction until the final documentation, is organized and consistent in the project development.

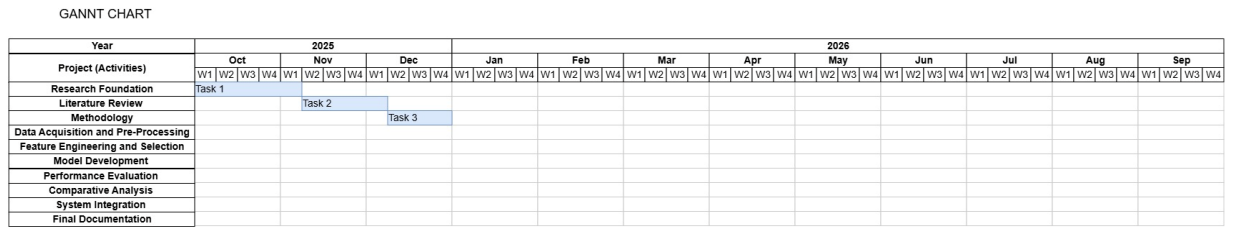


Figure 3.3 Gantt Chart

3.7 Summary

This chapter explains the research method used to build the proposed phishing detection system. The study followed the Waterfall model to maintain a structured and systematic development process, covering requirement analysis, system design, implementation, evaluation, and verification. The method also outlined the overall workflow, including dataset collection, data preprocessing, feature extraction with TF-IDF, and feature selection using Binary Harris Hawk Optimization (BHHO). The use of BHHO as a feature selection method forms the main contribution of this study, helping to manage the high dimensionality of TF-IDF features and improve classification efficiency. The improved feature set was then split into training and testing datasets, and classification was carried out using the Random Forest algorithm. System performance was assessed with standard measures such as accuracy, precision, recall, and the F1-score. This methodological framework creates a strong basis for the experimental analysis and the result of the proposed system will be explained and analyzed in the next chapter.

REFERENCES

- Ahmad, S., Zaman, M., Sami Al-Shamayleh, A., Ahmad, R., Muhammad Abdulhamid, S., Ergen, I., Akhunzada, A. (2025). Across the Spectrum In-Depth Review AI-Based Models for Phishing Detection. <https://doi.org/10.1109/OJCOMS.2024.3462503>
- Alshingiti, Z., Alaqel, R., Al-Muhtadi, J., Haq, Q. E. U., Saleem, K., & Faheem, M. H. (2023). A Deep Learning-Based Phishing Detection System Using CNN, LSTM, and LSTM-CNN. *Electronics*, 12(1), 232. <https://doi.org/10.3390/electronics12010232>
- Chanis, I., & Arampatzis, A. (2024). Enhancing phishing email detection with stylometric features and classifier stacking. *International Journal of Information Security*. <https://doi.org/10.1007/s10207-024-00928-7>
- Danasingh, A. A. G. S., Subramanian, A. B., & Epiphany, J. L. (2020). Identifying redundant features using unsupervised learning for high-dimensional data. *SN Applied Sciences*, 2(8), 1-13. <https://doi.org/10.1007/s42452-020-3157-6>
- GeeksforGeeks. (2025, July 23). *UCI Machine Learning Repository*. <https://www.geeksforgeeks.org/machine-learning/uci-machine-learning-repository/>
- Gualberto, E. S., de Sousa, R. T., Vieira, T. P. d. B., da Costa, J. P. C. L., & Duque, C. G. (2020). The Answer Is in the Text: Multi-Stage Methods for Phishing Detection Based on Feature Engineering. *IEEE Access*, 8, 213534-213547. <https://doi.org/10.1109/ACCESS.2020.3043396>

Jawade, J. V., & Ghosh, S. N. (2021). Phishing Website Detection Using Fast.ai library.

<https://doi.org/10.1109/ICCICT50803.2021.9510059>.

Kadulkar, S., Sherman, Z. M., Ganesan, V., & Truskett, T. M. (2022). Machine Learning–Assisted Design of Material Properties. *Annual Review of Chemical and Biomolecular Engineering*, 13, 1–25.

<https://doi.org/10.1146/annurev-chembioeng-092220-024340>

Kaur, A., Guleria, K., & Trivedi, N. K. (2021). *Feature selection in machine learning: Methods and comparison*. 2021 International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE), 118–122.

<https://doi.org/10.1109/ICACITE51222.2021.9404623>

Kustiawan, Y. A., & Ghauth, K. I. (2025). Evaluating the Impact of Feature Engineering in Phishing URL Detection: A Comparative Study of URL, HTML, and Derived Features. *IEEE Access*, 13, 1–18.

<https://doi.org/10.1109/ACCESS.2025.3579223>

Mandal, M., Singh, P. K., Ijaz, M. F., Shafi, J., & Sarkar, R. (2021). A Tri-Stage Wrapper-Filter Feature Selection Framework for Disease Classification. *Sensors*, 21(16), 5571.

<https://doi.org/10.3390/s21165571>

Mishra, S., & Soni, D. (2020). Smishing Detector: A security model to detect smishing through SMS content analysis and URL behavior analysis. *Future Generation Computer Systems*, 108, 582–596.

<https://doi.org/10.1016/j.future.2020.03.021>

Mohamad Asraf Daniel, Chong, S.-C., Chong, L.-Y., & Wee, K.-K. (2025). Optimising Phishing Detection: A Comparative Analysis of Machine Learning Methods with Feature Selection. *Journal of Informatics and Web Engineering*, 4(1), 200–212.

<https://doi.org/10.33093/jiwe.2025.4.1.15>

Mohammed, M. A., & El-Feky, S. F. (2025). *An enhanced filter-based approach for feature selection*. 2025 International Conference on Intelligent Methods, Systems and Applications (IMSA), 1–6. <https://doi.org/10.1109/IMSA65733.2025.11167724>

Mughaid, A., AlZu'bi, S., Hnaif, A., Taamneh, S., Alnajjar, A., & Abu Elsoud, E. (2022). An intelligent cyber security phishing detection system using deep learning techniques. *Cluster Computing*, 25, 3819–3828. <https://doi.org/10.1007/s10586-022-03604-4>

Saranya, G., Yamini Priya M. R., Kumaran K., Yeshwanth A. S., Aswathraj V., & Durai Nagendran M. (2025). AI-Powered Phishing Detection: A Data-Driven Cybersecurity Approach. <https://doi.org/10.1109/ICDSAAI65575.2025.11011572>

Savyanavar, A. S., Sankpal, P., & Mhala, N. C. (2024). Phishing Webpage Detection using Feature Selection Methods. *Journal of Electrical Systems*, 20(5s), 447–452. <https://doi.org/10.52783/jes.2070>

Siva, N., Sivaiah, B. V., Kishore Reddy, S. S., Irfan, S., Kumar, U. P., & Nayagara, S. N. (2024). Phishing Detection System through Hybrid Machine Learning Based on URL. *2024 5th International Conference for Emerging Technology (INCET)*, 1–6. <https://doi.org/10.1109/INCET61516.2024.10593373>

Syeda, D. Z., & Asghar, M. N. (2023). Enhancing Malware Classification: A Comparison of Filter and Embedded Feature Selection with a Novel Dataset. *2023 International Conference on Cyber Resilience (Cyber-RCI)*. <https://doi.org/10.1109/Cyber-RCI59474.2023.10671445>

Yin, Y., Jang-Jaccard, J., Xu, W., Singh, A., Zhu, J., Sabrina, F., & Kwak, J. (2023). IGRF-RFE: a hybrid feature selection method for MLP-based network intrusion detection on UNSW-NB15 dataset.

Journal of Big Data, 10(1), 1–15. <https://doi.org/10.1186/s40537-023-00694-8>

Zhang, H., Shi, Y., Liu, M., Chen, L., Wu, S., & Xue, Z. (2025). A combined feature selection approach for malicious email detection based on a comprehensive email dataset. *Cybersecurity*, 8(1), 14. <https://doi.org/10.1186/s42400-024-00309-6>

Kavya, S., & Sumathi, D. (2024). Design of a hybrid AI-based phishing website detection using LSTM, CNN, and Random Forest based ensemble learning analysis. In *2024 2nd International Conference on Networking and Communications (ICNWC)* (pp. 1–6). IEEE. <https://doi.org/10.1109/ICNWC60771.2024.10537482>

Yang, R., Zheng, K., Wu, B., Wu, C., & Wang, X. (2021). Phishing website detection based on deep convolutional neural network and random forest ensemble learning. *Sensors*, 21(24), Article 8281. <https://doi.org/10.3390/s21248281>

Keerthana, B. P., Siva, A., Senthilkumar, T., Palanisamy, T., Prabhu, N., & Srinivasan, K. (2025). Web phishing detection using decision tree random forest and XGBoost. In *Proceedings of the International Conference on Inventive Computation Technologies (ICICT-2025)*. IEEE. <https://doi.org/10.1109/ICICT64420.2025.11005064>

Gu, J., & Xu, H. (2022). An ensemble method for phishing websites detection based on XGBoost. In *2022 IEEE 14th International Conference on Computer Research and Development (ICCRD)* (pp. 195–199). IEEE. <https://doi.org/10.1109/ICCRD54409.2022.9730579>

Saraswathi, P., Keerthi, S., Reshmika, K. S., Pritham, G. R. G., Prabha, M., & Dhiyanesh, B. (2024). Evaluating the efficacy of machine learning methods in phishing detection: A comparative analysis. In *2024 International Conference on Big Data and Smart*

Computing (ICBDS). IEEE.

<https://doi.org/10.1109/ICBDS61829.2024.10837224>

Kalikiri, J., Policepatel, P. R., Kancharla, S., Endurthy, R. R., Kotteti, C. M. M., & Lal, R. (2025). URL-based phishing attacks detection using machine learning. In *2025 International Conference on Networks, Cryptography and Information Security (NetCrypt)*. IEEE.

<https://doi.org/10.1109/NETCRYPT65877.2025.11102732>

Ajibade, S. M. M., Jasser, M. B., Alebiosu, D. O., Issa, B., San, L. W., & ALDharhani, G. S. (2025). A comparative analysis of detection methods for phishing websites using machine learning. In *2025 IEEE 10th International Conference on Computing, Asia-Pacific Region (I2CACIS)*. IEEE.

<https://doi.org/10.1109/I2CACIS65476.2025.11100418>

Joseph, A. S. K., & Srinivasan, S. (2025). Anti-phishing adaptive AI systems: Efficiently countering social engineering attacks by real-time analysis of email content. In *2025 International Conference on Computational Innovations and Engineering Sustainability (ICCIES)*. IEEE. <https://doi.org/10.1109/ICCIES63851.2025.11032758>

Doshi, T., Patel, V., Shah, N., Swain, D., Swain, D., & Acharya, B. (2025). PhishHunter-XLD: An ensemble approach integrating machine learning and deep learning for phishing URL classification. *Franklin Open*, 12, Article 100349.

<https://doi.org/10.1016/j.fraope.2025.100349>